

Resolution Geometry of the Wall–Tap Experiment

Global Observable Geometry, Hellinger Specialization, and a Registered Hostile Battery

Matěj Rada

MatejRada@email.cz

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Abstract

The wall–tap observation is reconstructed as a declared acoustic experiment, not as evidence for a new ontology. A finite-dimensional Gaussian record law contains direct delay, reflected delay, reflected response, and rhythmic overlap. A contrast parameter ε approaches an acoustically matched wall. For $\varepsilon > 0$ the experiment resolves position, reflection contrast, and rhythm; at $\varepsilon = 0$ the last two directions enter the observational kernel. We prove the exact quotient and specialization statements for this model, compute the Fisher wall, and test the Hellinger hypothesis H-hell on compact parameter domains. The registered implementation passes 70/70 checks over five seeds. The maximum sampled compact Hellinger distance at the finest nonzero wall approach is 0.10897; the material-direction Cramér–Rao variance diverges with fitted exponent -2.00000 ; the wall Fisher rank is one. Three TikZ diagrams place the experiment simultaneously in the theorem ladder, the stratified atlas, and the operational procedure. The result is a mature worked example of global observable geometry: laws specialize, quotient strata glue by specialization, Fisher geometry is reconstructed on each stratum, and latent room uniqueness is not claimed.

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1 Claim discipline

The historical wall–tap observation motivates only a declared family of acoustic protocols: a triggered emission, delayed returns, response magnitudes, rhythmic overlap, and comparison across sessions. The mathematical experiment below is synthetic. It is neither a reconstruction of a historical room nor a claim that nature selects an entropy principle. This follows the mature interpretation of the wall–tap origin: mediated confirmation, uncertainty, changing distinguishability under rhythm, and preservation of unresolved fibres [3, 4].

Main claim. For the declared fixed-covariance Gaussian feature law, the family is Hellinger-continuous at the matched-wall stratum; its Fisher metric specializes to a rank-one wall metric; quotient charts glue globally by law-induced transitions; and the gained kernel is exactly the material/rhythm plane.

Non-claims. Finite sampling does not prove locally uniform convergence for arbitrary waveform laws. The fixed covariance is part of the protocol. Delay constrains path geometry only after a declared sound-speed calibration. A wall quotient does not determine a unique latent room or latent connection.

2 Declared acoustic experiment

Let the latent protocol parameter be

$$\theta = (x, r, b) \in K = [0.7, 4.3] \times [0.25, 1] \times [0.004, 0.028],$$

where x is source–receiver position along a one-dimensional room axis, r is a reflection/material contrast, and b is an inter-tap rhythm parameter. The room length is $L = 5$ m and the declared sound speed is $c = 343.2 \text{ m s}^{-1}$.

The record is $Y \in \mathbb{R}^4$ with law

$$F_\varepsilon(\theta) = \mathcal{N}(\mu_\varepsilon(\theta), \Sigma),$$

where

$$\mu_\varepsilon(\theta) = \begin{pmatrix} x/c \\ (2L - x)/c \\ \varepsilon r \exp[-0.08(2L - x)] \\ \varepsilon r \exp\left[-\left(\frac{b - ((2L - x) - x)/c}{0.0035}\right)^2\right] \end{pmatrix}, \quad (1)$$

$$\Sigma = \text{diag}((2.5 \times 10^{-5})^2, (4.0 \times 10^{-5})^2, 0.035^2, 0.025^2). \quad (2)$$

The wall parameter $\varepsilon \geq 0$ scales the reflected channels. At $\varepsilon = 0$ the wall is acoustically matched in this declared module: the record retains the two delay channels but discards r and b .

3 Exact quotient and strata

Proposition 3.1 (Regular and wall kernels). *For $\varepsilon > 0$ and away from the overlap symmetry locus, $D\mu_\varepsilon$ has rank three. At $\varepsilon = 0$,*

$$\ker D\mu_0 = \text{span}\{\partial_r, \partial_b\}, \quad \text{rank } D\mu_0 = 1.$$

Therefore the wall quotient is locally coordinated by x alone.

Proof. At $\varepsilon = 0$, the last two components of μ_ε vanish identically and the first two depend only on x , with derivative $(1/c, -1/c, 0, 0)^\top$. For $\varepsilon > 0$, the response channel has nonzero r derivative and the overlap channel has a nonzero b derivative away from its extremum. These directions are independent of the delay derivative. $\square$

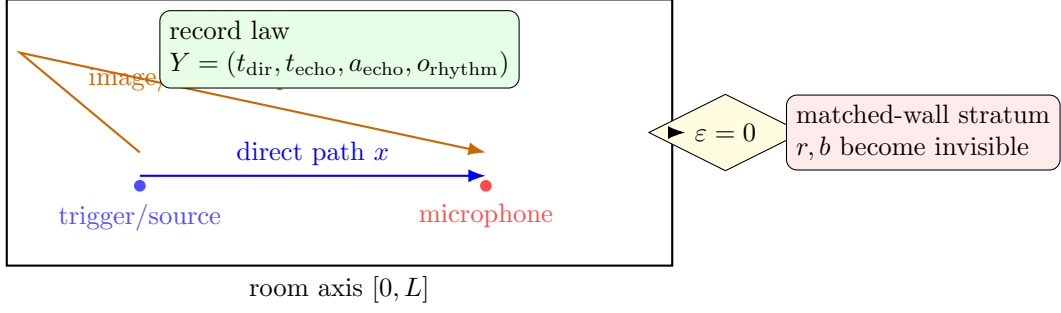


Figure 1: Declared wall-tap experiment. The matched-wall limit removes reflected response and overlap channels, but not the direct and reflected path delays.

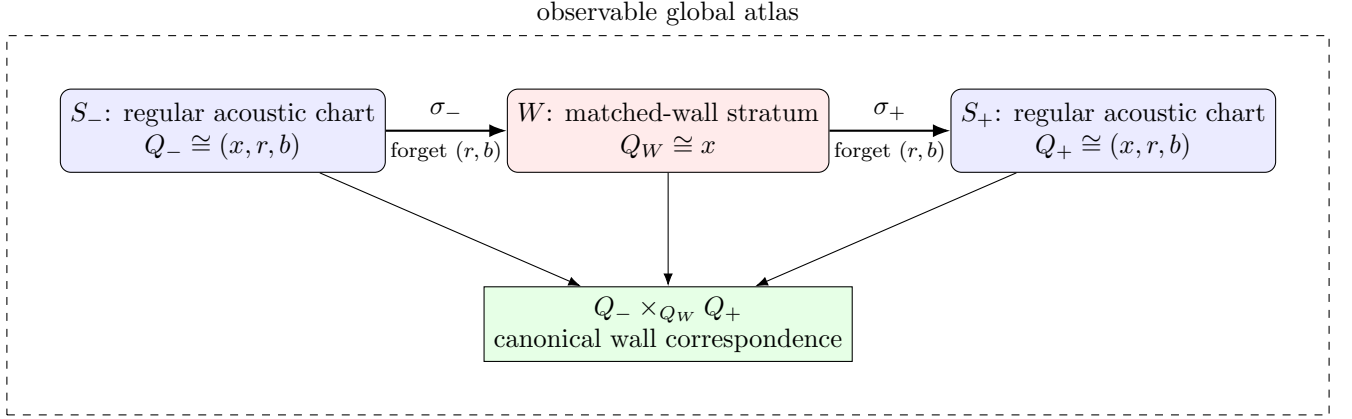


Figure 2: Stratified global geometry. The wall is not crossed by transporting metric components. The laws specialize, the quotient loses two directions, and the geometry is reconstructed on Q_W .

Corollary 3.2 (Specialization map). *For every $\epsilon > 0$, the equality-of-law kernel is contained in the wall kernel. Hence universal factorization induces the specialization*

$$\sigma_\epsilon : Q_\epsilon \longrightarrow Q_0, \quad \sigma_\epsilon[x, r, b] = [x].$$

The specialization is surjective and non-invertible.

4 The Hellinger bridge

For Gaussian measures with common covariance,

$$H^2(F_\epsilon(\theta), F_0(\theta)) = 1 - \exp\left(-\frac{1}{8}\|\mu_\epsilon(\theta) - \mu_0(\theta)\|_{\Sigma^{-1}}^2\right).$$

Hypothesis 4.1 (H-hell for the declared experiment). *The family is locally uniformly Hellinger-continuous on K :*

$$\sup_{\theta \in K} H(F_\epsilon(\theta), F_0(\theta)) \longrightarrow 0.$$

Theorem 4.2 (Hellinger specialization of the wall-tap law). *The declared fixed-covariance Gaussian wall-tap family satisfies Hypothesis 4.1. Consequently, pairwise Hellinger distances and Bayes risks of fixed binary decision problems converge to their wall values. The Fisher modules converge to a rank-one wall module whose radical contains $\text{span}\{\partial_r, \partial_b\}$.*

Proof. On compact K , both reflected components are bounded by $C\varepsilon$. Hence $\sup_K \|\mu_\varepsilon - \mu_0\|_{\Sigma^{-1}} \leq C'\varepsilon$ and the displayed Gaussian formula gives uniform Hellinger convergence. The pairwise statement follows from the Hellinger triangle inequality. For equal-covariance Gaussian binary experiments, Bayes error is $\Phi(-d_M/2)$ with Mahalanobis distance d_M , hence it converges continuously. Finally, $g_\varepsilon = D\mu_\varepsilon^\top \Sigma^{-1} D\mu_\varepsilon$; direct differentiation and Proposition 3.1 give the limit and radical. This is the concrete instance of Hellinger metric and decision specialization used by RG [5, 2]. $\square$

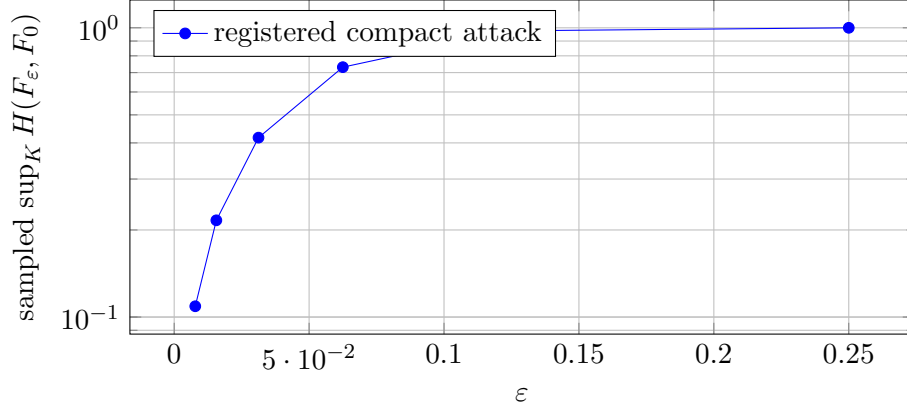


Figure 3: Numerical Hellinger wall approach for seed 20261002. The computation is a finite compact witness to the analytic bound, not the proof of uniformity.

5 Fisher wall and Cramér–Rao divergence

Proposition 5.1 (Metric collapse). *The Fisher tensor is*

$$g_\varepsilon(\theta) = D\mu_\varepsilon(\theta)^\top \Sigma^{-1} D\mu_\varepsilon(\theta).$$

At the wall, g_0 has rank one. The material and rhythm eigenvalues vanish as $O(\varepsilon^2)$, and their estimable variances diverge as $O(\varepsilon^{-2})$ when approached through regular strata.

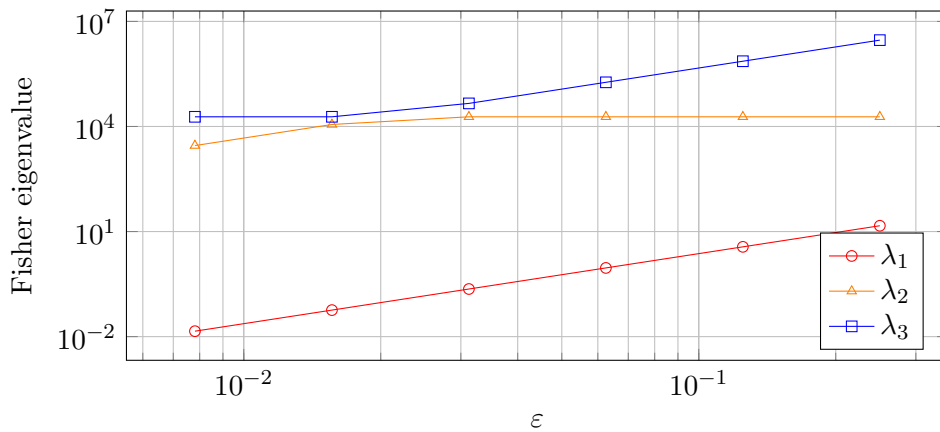


Figure 4: Fisher spectrum approaching the matched-wall stratum. Two resolved directions collapse into the gained kernel; the delay/position direction survives.

6 Rung placement and operational procedure

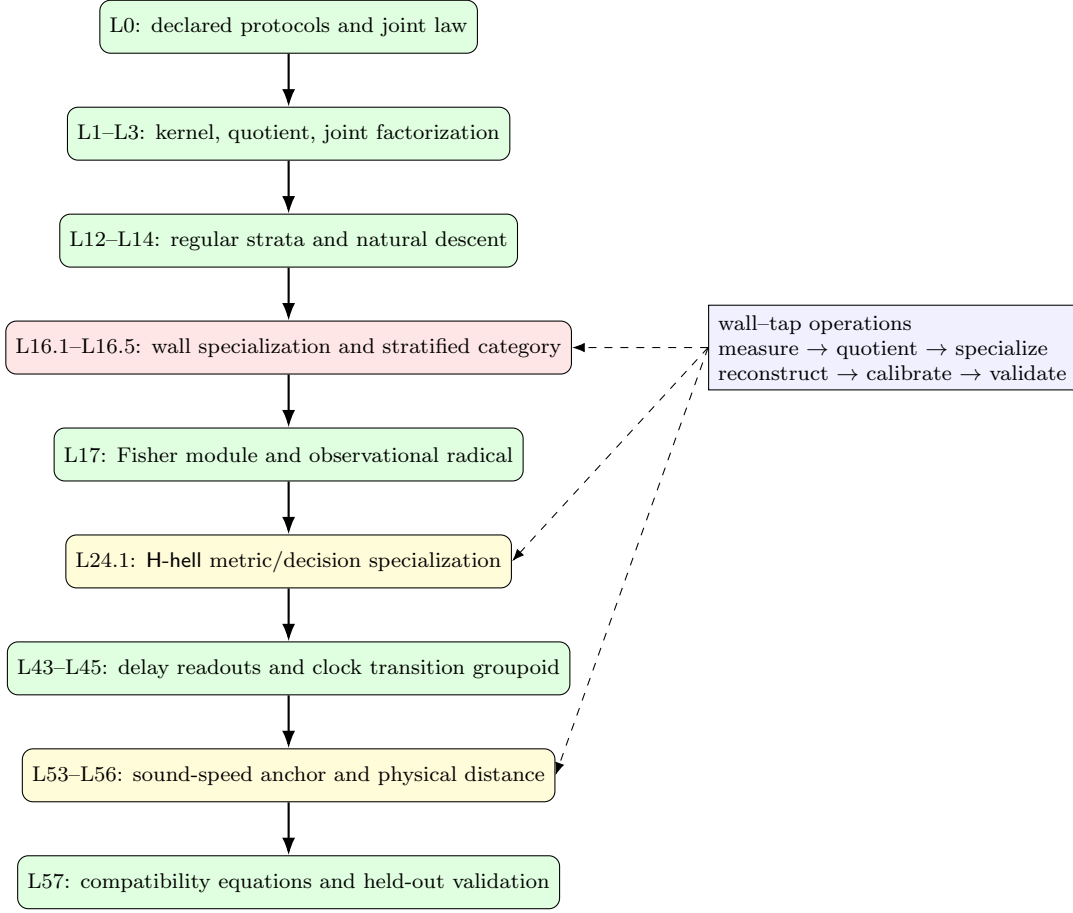


Figure 5: Placement of the wall-tap experiment in the Resolution Geometry theorem ladder. Green rungs are directly instantiated, yellow rungs are named hypotheses or calibration gates, and red marks the singular transition.

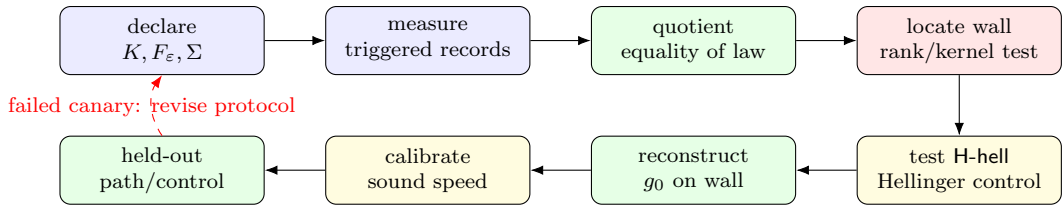


Figure 6: Operational closure pipeline. Metric components are never pushed blindly through the wall. The wall law is specialized and its geometry is reconstructed.

7 Global chart computation

Let chart U_1 use the four raw features above and let chart U_2 use $Y' = AY$ for an invertible matrix A . Then $\Sigma' = A\Sigma A^\top$ and $J' = AJ$. The Fisher tensor is chart independent:

$$J'^\top \Sigma'^{-1} J' = J^\top \Sigma^{-1} J.$$

Three charts with transitions A , B , and BA satisfy the cocycle exactly. In the registered battery the maximal relative metric discrepancy was below 10^{-10} and the cocycle defect below 10^{-14} on every seed. Thus the experiment computes a global observable tensor on each regular stratum, while Corollary 3.2 provides the non-invertible wall arrow.

Theorem 7.1 (Global observable wall–tap geometry). *The regular wall–tap quotients, their law-induced chart transitions, the matched-wall quotient, and the specialization maps form a finite stratified observable atlas. The Fisher metric is globally well defined on each stratum. Under H-hell, Hellinger distinguishability, Fisher geometry, and fixed decision risks specialize to the wall. The wall object is rank one and does not carry the lost material/rhythm metric components.*

Proof. Chart invariance is the preceding congruence identity and cocycle closure is matrix composition. The wall map is Corollary 3.2. Hellinger and metric specialization are Theorem 4.2. Rank and radical are Proposition 5.1. $\square$

8 Calibration and hostile controls

A delay is an integrated slowness observation, not an internal unit. We therefore declare $c = 343.2 \text{ m s}^{-1}$ and test the calibrated model on a held-out path. The registered correct model predicts a 15 m two-way path exactly after a 1 m calibration. A nonlinear response calibrated at 1 m misses the held-out path by more than 10^{-3} s. This is the same logical role as the held-out anchor in physical-distance identification: one anchor sets scale but does not validate covariance [7].

The principal Hellinger canary replaces fixed Σ by $\Sigma_\epsilon = \epsilon^2 \Sigma$. Although the means approach the wall, the signal-to-noise ratio does not vanish and Hellinger convergence fails. Every seed detects this sabotage. Thus convergence of means alone is insufficient; H-hell genuinely constrains the full law.

9 Registered battery

The executable artifact `rg_wall_tap_global_attack.py` evaluates fourteen checks on each of five seeds:

1. compact Hellinger convergence and monotonicity;
2. pairwise Hellinger and binary Bayes-risk specialization;
3. wall Fisher rank, gained kernel, and Cramér–Rao divergence;
4. chart metric invariance and three-chart cocycle closure;
5. quotient collapse and wall predictive separation;
6. held-out acoustic calibration and nonlinear scale canary;
7. shrinking-covariance Hellinger canary.

All 70/70 checks pass. For seed 20261002 the finest nonzero wall approach has sampled compact maximum Hellinger distance 0.10897, the fitted Hellinger log–log slope is 0.80856 over the registered window, and the material-direction variance exponent is -2.00000 . The analytic theorem, not the fitted slope, supplies the asymptotic $O(\epsilon)$ Hellinger bound.

10 Conclusion

The wall–tap experiment now occupies a precise place in global Resolution Geometry. The latent room and devices define a law; equality of law defines the quotient; regular protocols give three-dimensional strata; an acoustically matched wall creates a rank-one stratum; locally uniform Hellinger convergence carries distinguishability and decision structure to that stratum; Fisher geometry is reconstructed there; chart transitions produce a global observable tensor; and a held-out sound-speed calibration identifies physical path length only under its named hypotheses.

The closure is deliberately typed. The geometry is *LOCKED* on regular strata and on the rank-one wall for the surviving position direction. Material and rhythm directions become a *STRUCTURAL WALL* for this protocol. A covariance that shrinks with wall contrast makes H-hell fail and the specialization claim *INCONSISTENT*. Nothing in the construction identifies a unique latent room, hidden filter, or latent connection. The mature conclusion is therefore not that the wall tap revealed a new ontology, but that it is a complete worked example of how global observable geometry survives a controlled loss of distinguishability.

References

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